

Index

(Euclidean) length,	Item				
1-norm (vector),	Item				
1-norm, vector,	Definition				
2-norm (vector),	Item				
2-norm, matrix,	Definition				
2-norm, vector,	Definition				
$[\cdot]$,	Remark				
ϵ_{mach} ,	Paragraph				
$\text{fl}(\cdot)$,	Paragraph				
γ_n ,	Definition	Definition			
∞ -norm (vector),	Item				
∞ -norm, vector,	Definition				
$\kappa(A)$,	Definition	Definition			
$\text{maxi}(\cdot)$,	Paragraph				
$\overline{A}$,	Definition				
$\overline{x}$,	Paragraph				
$\cdot$,	Paragraph				
θ_j ,	Lemma	Lemma			
$ \cdot $,	Paragraph				
e_j ,	Paragraph				
I ,	Paragraph				
p -norm (vector),	Item				
p -norm, matrix,	Definition				
p -norm, vector,	Definition				
absolute value,	Subsection	Paragraph	Item		
ACM,	Subsubsection	Paragraph			
Alternative Computational Model,	Subsubsection	Paragraph			
axpy,	Paragraph				
backward stable implementation,	Definition				
Basic Linear Algebra Subprograms,	Subsection	Remark			
BLAS,	Subsection	Remark			
blocked algorithm,	Paragraph				
catastrophic cancellation,	Section				
Cauchy-Schwarz inequality,	Paragraph	Theorem			
CGS,	Subsection				
characteristic polynomial,	Subsection	Definition			
chasing the bulge,	Paragraph				
Cholesky decomposition,	Week				
Cholesky factor,	Paragraph				
Cholesky factorization,	Paragraph	Week			
Cholesky factorization theorem,	Theorem	Theorem			
Classical Gram-Schmidt,	Subsection				
complex conjugate,	Paragraph				
complex product,	Item				
condition number,	Definition	Definition	Definition	Definition	
conjugate,	Paragraph	Item			
conjugate (of matrix),	Item				
conjugate (of vector),	Item				
conjugate of a matrix,	Definition				
conjugate transpose (of matrix),	Item				
conjugate transpose (of vector),	Item				
consistent matrix norm,	Definition	Definition	Definition	Definition	
cost of basic linear algebra operations,	Section				
cubic convergence,	Item	Item			
defective matrix,	Paragraph	Definition	Definition		
deflation,	Subsection				
descent methods,	Week				
determinant,	Paragraph				
direction of maximal magnification,	Item				
distance,	Paragraph				
dot product,	Item	Definition			
eigenpair,	Definition	Definition			
eigenvalue,	Definition	Definition			
eigenvector,	Definition	Definition			
elementary elementary pivot matrix,	Paragraph				
equivalence style proof,	Homework				
Euclidean distance,	Paragraph				
exact descent method,	Paragraph				
fill-in,	Paragraph				
fixed-point equation,	Paragraph				
FLAME notation,	Remark				
floating point numbers,	Subsection				

flop, [Item](#) [Item](#) [Item](#)
forward substitution, [Paragraph](#)
Frobenius norm, [Definition](#) [Item](#)

Gauss transform, [Subsection](#)
Gaussian elimination, [Section](#) [Subsection](#)
Gaussian elimination with row exchanges, [Subsection](#)
Geometric multiplicity, [Definition](#) [Definition](#)
Givens' rotation, [Subsection](#)
gradient, [Paragraph](#)
Gram-Schmidt orthogonalization, [Section](#)

Hermitian, [Definition](#)
Hermitian Positive Definite, [Item](#) [Subsection](#)
Hermitian positive definite, [Definition](#)
Hermitian transpose, [Remark](#) [Definition](#)
Hermitian transpose (of matrix), [Item](#)
Hermitian transpose (of vector), [Item](#)
Hessenberg matrix, [Definition](#)
homogeneity (of absolute value), [Item](#)
homogeneity (of matrix norm), [Item](#)
homogeneity (of vector norm), [Item](#) [Item](#)
homogeneneity (of matrix norm), [Item](#)
Householder reflector, [Definition](#) [Definition](#)
Householder transformation, [Paragraph](#) [Definition](#) [Definition](#) [Definition](#)
Housev, [Paragraph](#) [Paragraph](#) [Paragraph](#)
Housev1, [Paragraph](#)
HPD, [Item](#) [Subsection](#) [Definition](#)

identity matrix, [Paragraph](#)
Implicit Q Theorem, [Theorem](#) [Theorem](#)
induced matrix norm, [Subsection](#) [Definition](#)
infinity norm, [Paragraph](#)
inner product, [Item](#) [Definition](#)

Jordan Canonical Form, [Subsection](#)

Krylov subspace, [Definition](#) [Definition](#)

left pseudo inverse, [Definition](#)
left pseudo-inverse, [Paragraph](#)
left singular vector, [Theorem](#) [Theorem](#)
Legendre polynomials, [Paragraph](#)
linear convergence, [Definition](#) [Item](#) [Definition](#) [Item](#)
linear least squares, [Week](#)
linear transformation, [Subsection](#)
LLS, [Week](#)
LU decomposition, [Week](#) [Definition](#) [Paragraph](#) [Definition](#)
LU factorization, [Week](#) [Section](#) [Definition](#) [Paragraph](#) [Definition](#)
LU factorization - existence, [Theorem](#) [Theorem](#)
LU factorization algorithm (bordered), [Ponder This](#)
LU factorization algorithm (left-looking), [Paragraph](#)
LU factorization algorithm (right-looking), [Subsection](#)
LU factorization with complete pivoting, [Subsection](#)
LU factorization with partial pivoting, [Subsection](#)
LU factorization with partial pivoting (right-looking algorithm), [Subsection](#)
LU factorization with pivoting, [Section](#)

machine epsilon, [Paragraph](#) [Paragraph](#) [Definition](#) [Definition](#)
magnitude, [Paragraph](#)
matrix, [Subsection](#) [Paragraph](#)
matrix 1-norm, [Item](#)
matrix 2-norm, [Definition](#) [Item](#)
matrix ∞ -norm, [Item](#)
matrix p -norm, [Definition](#)
matrix norm, [Definition](#) [Definition](#)
matrix norm, 2-norm, [Definition](#)
matrix norm, p -norm, [Definition](#)
matrix norm, consistent, [Definition](#) [Definition](#) [Definition](#) [Definition](#)
matrix norm, Frobenius, [Definition](#)
matrix norm, induced, [Subsection](#) [Definition](#)
matrix norm, submultiplicative, [Subsection](#) [Definition](#) [Definition](#)
matrix norm, subordinate, [Definition](#) [Definition](#)
matrix p-norm, [Item](#)
matrix-vector multiplication, [Paragraph](#)
memop, [Item](#) [Item](#) [Item](#)
Method of Multiple Relatively Robust Representations (MRRR), [Subsection](#)
Method of Normal Equations, [Paragraph](#)
method of normal equations, [Section](#)
MRRR, [Subsection](#)

natural ordering, [Item](#)
nested dissection, [Paragraph](#)
norm, [Week](#)
norm, Frobenius, [Definition](#)
norm, infinity, [Paragraph](#)
norm, matrix, [Definition](#) [Definition](#)
norm, vector, [Definition](#) [Definition](#)
normal equations, [Section](#) [Item](#)

numerical stability, [Week](#)

orthogonal matrix, [Remark](#)

orthogonal projection, [Paragraph](#)

orthogonal vectors, [Definition](#)

orthonormal matrix, [Definition](#)

orthonormal vectors, [Definition](#)

over-relaxation, [Paragraph](#)

parent functions, [Item](#)

partial pivoting, [Paragraph](#) Subsection

pivot, [Paragraph](#)

pivot element, [Paragraph](#)

positive definite, [Definition](#)

positive definiteness (of absolute value), [Item](#)

positive definitenessx (of matrix norm), [Item](#) [Item](#)

positive definitenessx (of vector norm), [Item](#) [Item](#)

precondition, [Item](#)

principal leading submatrix, [Definition](#) [Definition](#)

pseudo inverse, [Definition](#) [Paragraph](#)

pseudo-inverse, [Paragraph](#)

QR algorithm, [Subsection](#)

QR decomposition, [Week](#)

QR Decomposition Theorem, [Theorem](#) [Theorem](#)

QR factorization, [Week](#)

QR factorization with column pivoting, [Subsection](#) [Paragraph](#)

quadratic convergence, [Item](#) [Item](#)

Rank Revealing QR, [Subsection](#)

rank-k update, [Item](#)

Rayleigh quotient, [Definition](#) [Definition](#)

Rayleigh Quotient Iteration, [Subsection](#)

reflector, [Paragraph](#) [Definition](#) [Definition](#)

residual, [Homework](#)

right pseudo inverse, [Ponder This](#)

right singular vector, [Theorem](#) [Theorem](#)

rotation, [Subsubsection](#)

rowl pivoting, [Paragraph](#)

RRQR, [Subsection](#)

Schur decomposition, [Subsection](#) [Theorem](#) [Theorem](#)

Schur Decomposition Theorem, [Theorem](#) [Theorem](#)

SCM, [Subsubsection](#) [Paragraph](#)

separator, [Paragraph](#)

shifted inverse power method, [Subsection](#)

shifted QR algorithm, [Subsection](#)

similarity transformation, [Definition](#) [Definition](#)

Singular Value Decomposition, [Week](#) [Paragraph](#)

singular vector, [Theorem](#) [Theorem](#)

solving triangular systems, [Subsection](#)

SOR, [Paragraph](#)

sparse linear system, [Week](#)

Spectral decomposition, [Subsection](#)

spectral decomposition, [Theorem](#) [Theorem](#)

Spectral Decomposition Theorem, [Theorem](#) [Theorem](#)

spectral radius, [Definition](#) [Definition](#)

spectrum, [Definition](#) [Definition](#) [Definition](#) [Definition](#)

stability, [Week](#)

standard basis vector, [Definition](#) [Definition](#) [Definition](#) [Definition](#)

Standard Computational Model, [Subsubsection](#) [Paragraph](#)

submultiplicative matrix norm, [Subsection](#) [Definition](#) [Definition](#)

subordinate matrix norm, [Definition](#) [Definition](#)

subspace iteration, [Subsection](#) [Paragraph](#)

successive over-relaxation, [Paragraph](#)

superlinear convergence, [Item](#) [Item](#)

superquadratic convergence, [Item](#)

SVD, [Week](#) [Paragraph](#)

symmetric positive definite, [Definition](#) [Paragraph](#) [Definition](#)

tall and skinny, [Paragraph](#)

The Francis implicit QR Step, [Subsection](#)

The implicit Q theorem, [Subsection](#)

transpose, [Definition](#)

transpose (of matrix), [Item](#)

transpose (of vector), [Item](#)

triangle inequality (for absolute value)), [Item](#)

triangle inequality (for matrix norms)), [Item](#) [Item](#)

triangle inequality (for vector norms)), [Item](#) [Item](#)

triangular solve with multiple right-hand sides, [Item](#)

triangular system, [Subsection](#)

TRSM, [Item](#)

unit ball, [Subsection](#) [Item](#)

unit roundoff, [Paragraph](#) [Definition](#) [Definition](#)

unit roundoff error, [Paragraph](#)

unitary matrix, [Definition](#) [Definition](#)

unitary similarity transformation, [Definition](#) [Definition](#)

upper Hessenberg matrix, [Definition](#)

Vandermonde matrix, [Paragraph](#)
vector 1-norm, [Definition](#) [Item](#)
vector 2-norm, [Definition](#) [Item](#)
vector ∞ -norm, [Item](#)
vector ∞ -norm, [Definition](#)
vector p -norm, [Item](#)
vector p -norm, [Definition](#)
vector norm, [Definition](#) [Definition](#)
vector norm, 1-norm, [Definition](#)
vector norm, 2-norm, [Definition](#)
vector norm, ∞ -norm, [Definition](#)
vector norm, p -norm, [Definition](#)

Wilkinson shift, [Item](#)